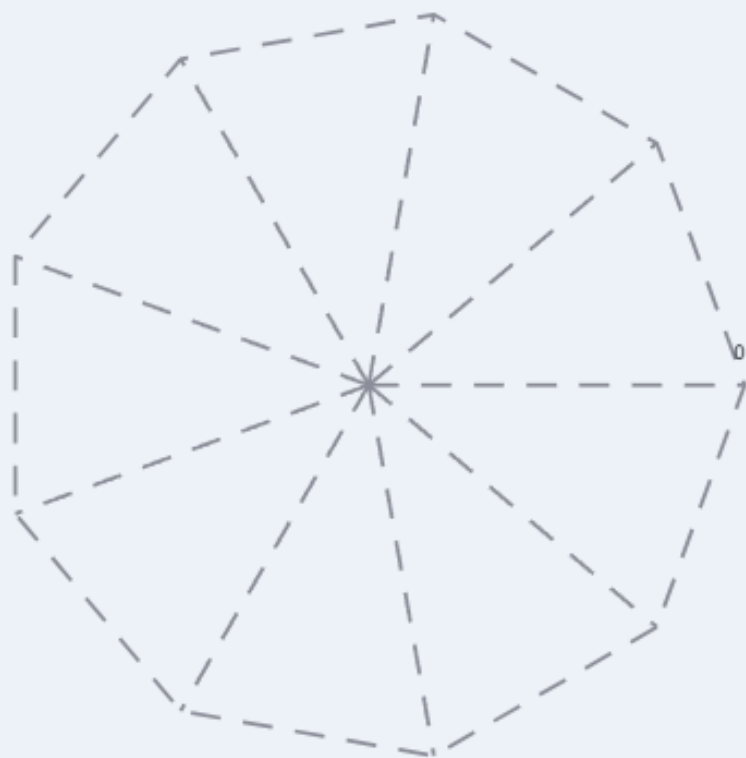




Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



- A 0% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 0% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 0% Length of Curves
- F 0% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult



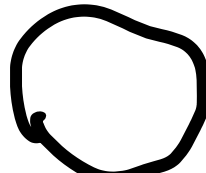
My method to graph polar coordinates

① Look at θ (ignore r first)
angle

↳ Plot a graph for θ

② Identify regions for $r \geq 0$ and $r < 0$

③ Draw the circles according to what is given by r .

Answer: The intersection of  and region for θ

Ex. 10.3 Q 18

$$\theta = 11\pi/4, \quad r \geq -1$$

495 degree

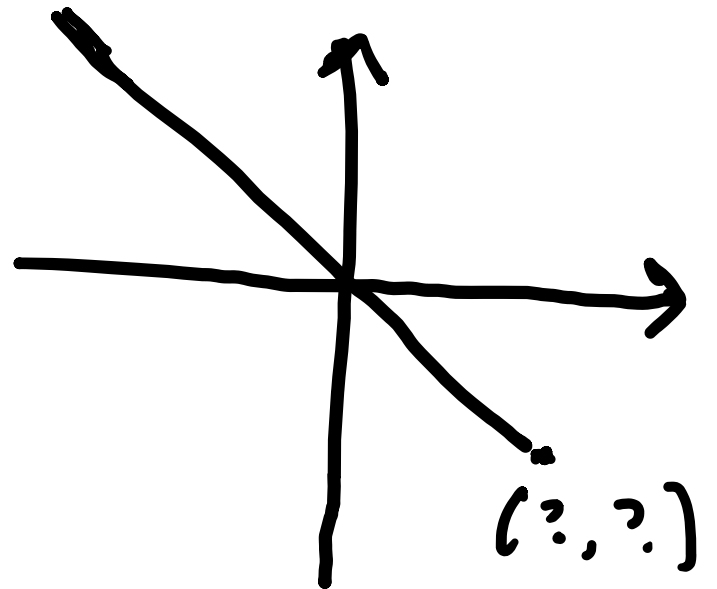
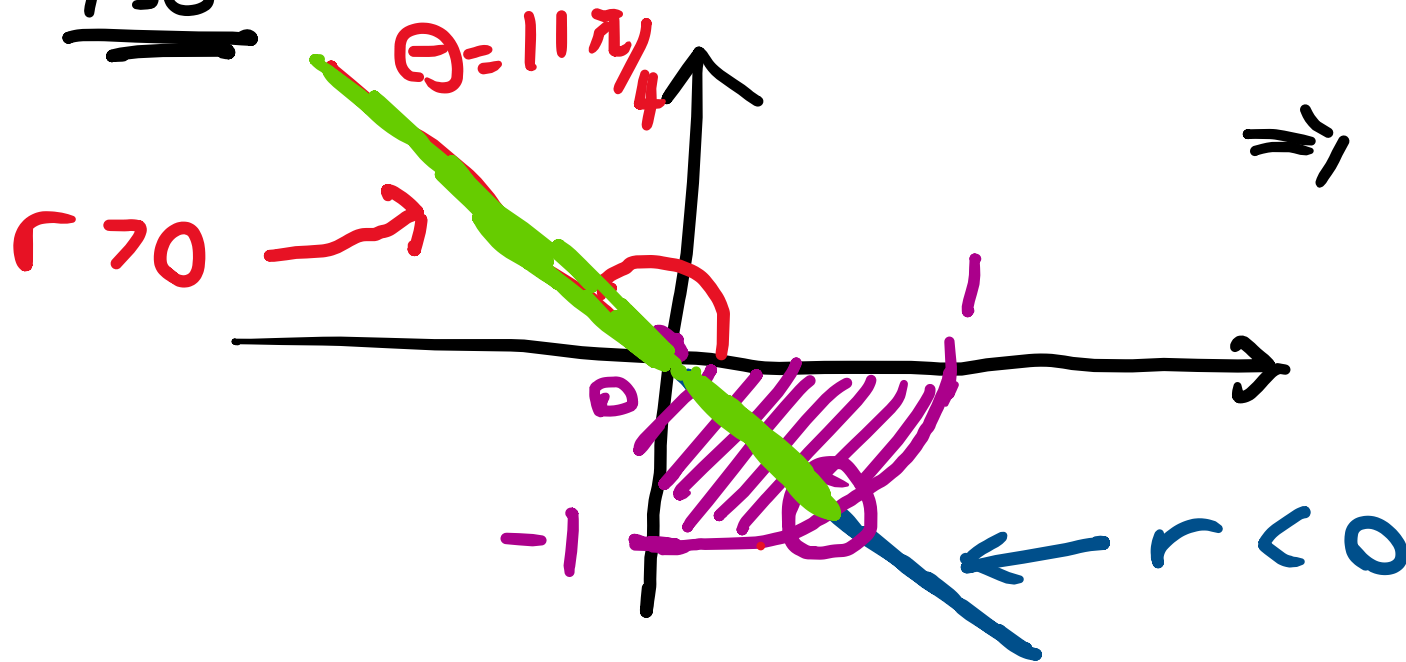
"

135°

$$-1 \leq r < 0 \text{ or } r \geq 0$$

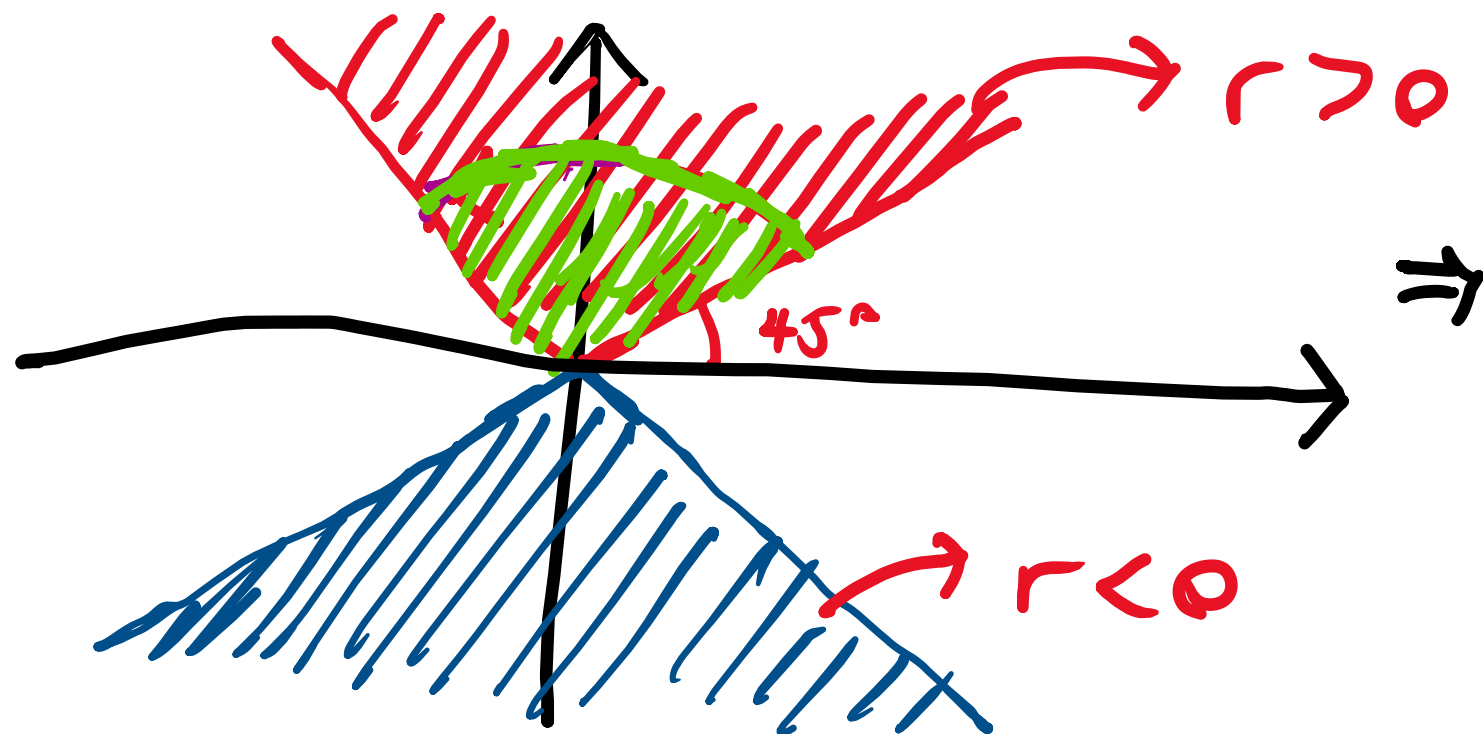
$$r \geq 0$$

Answer:



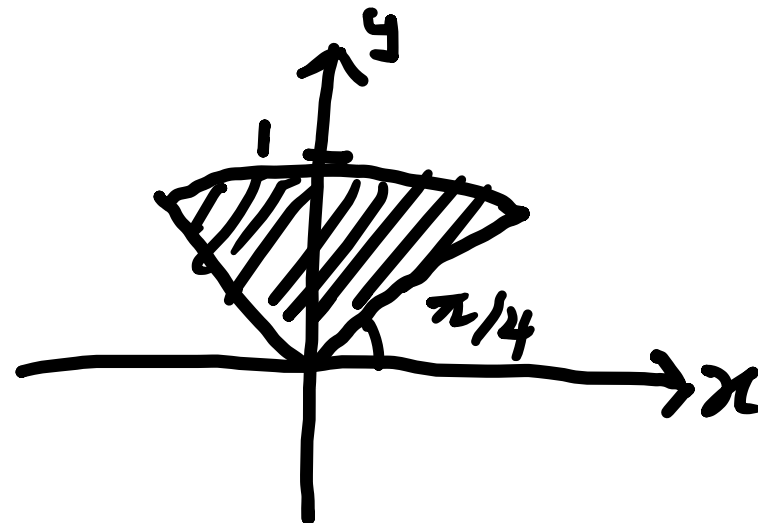
Q23

$$\underbrace{\pi/4}_{45^\circ} \leq \Theta \leq \underbrace{3\pi/4}_{135^\circ}, \quad \underbrace{0 \leq r \leq 1}_{\text{red wavy line}}$$



$\Rightarrow$

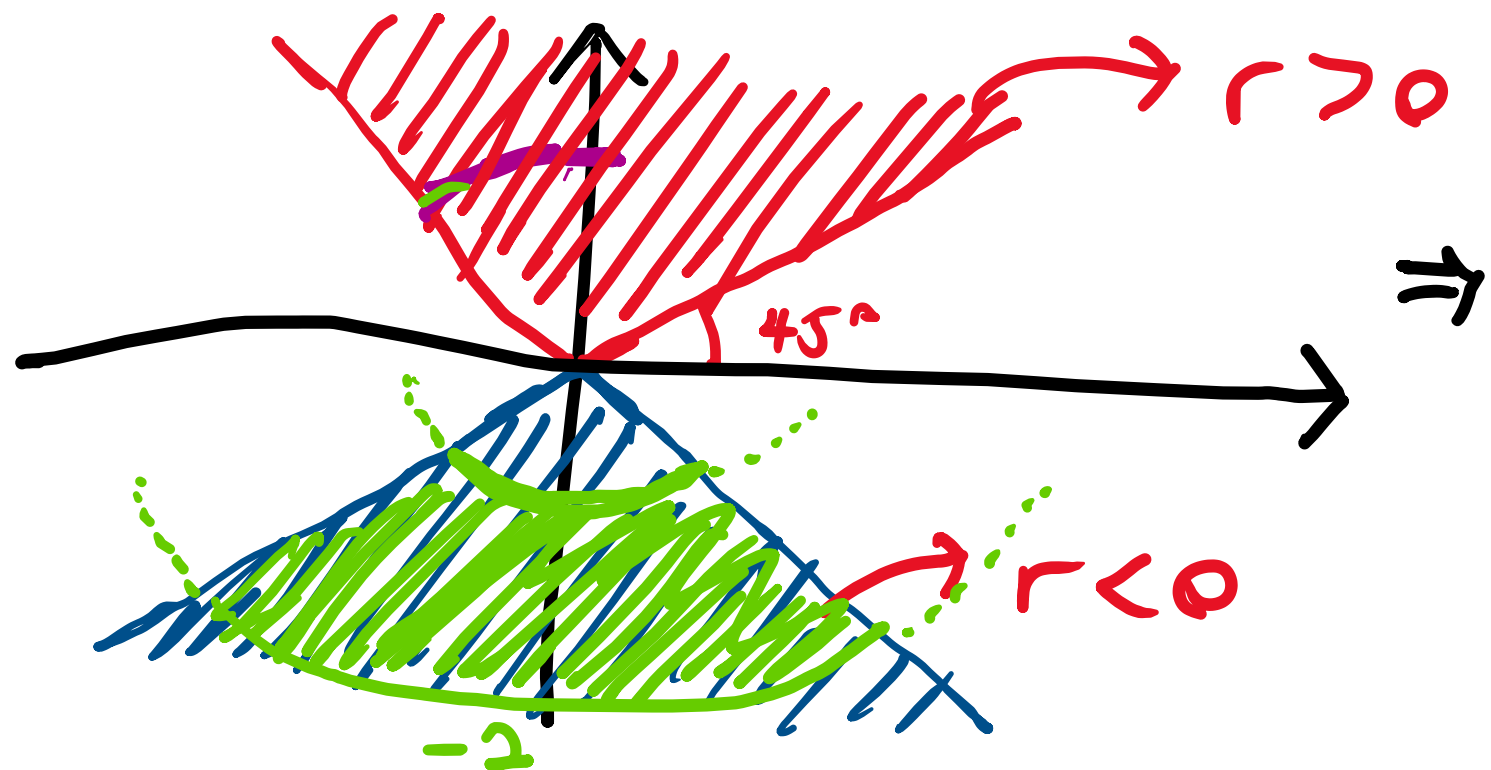
Ans:



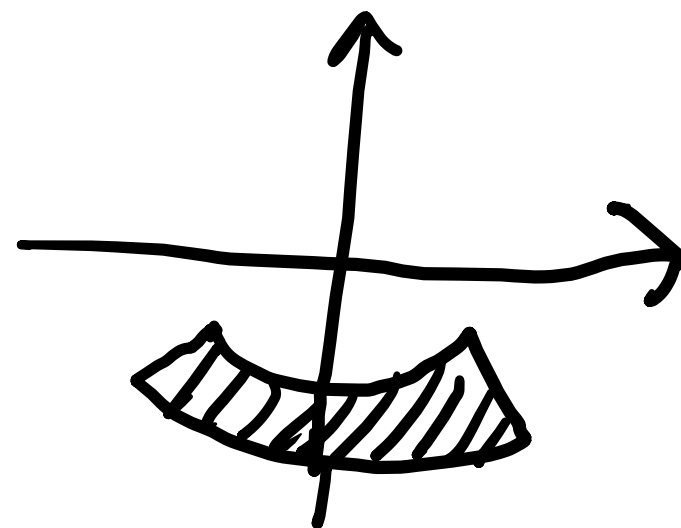
Q23

$$\underbrace{\pi/4}_{45^\circ} \leq \Theta \leq \underbrace{3\pi/4}_{135^\circ},$$

$$\boxed{-2 \leq r \leq -\frac{1}{2}}$$

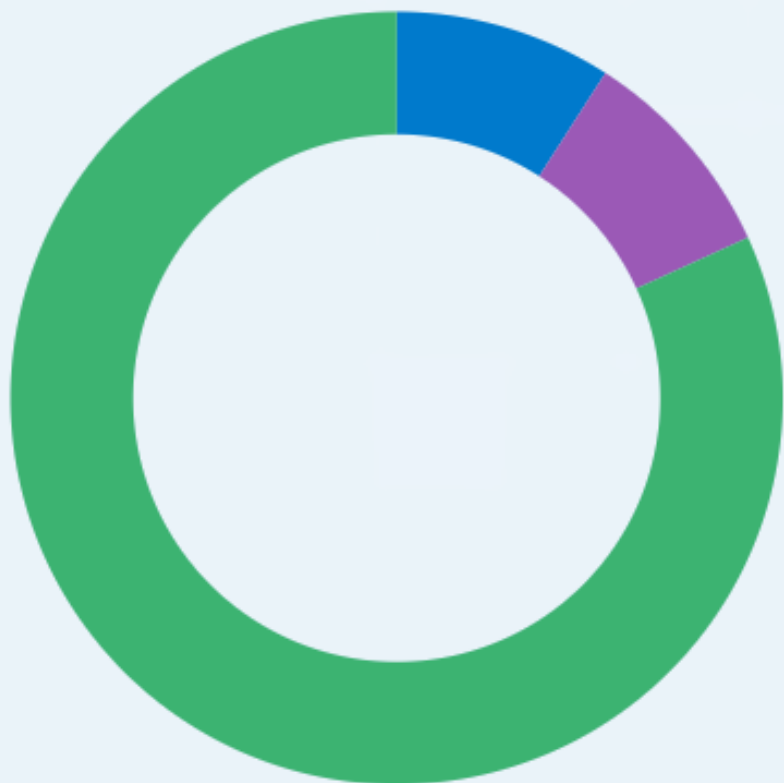


Ans:





Choose the correct graph for $0 \leq \theta \leq \pi/2$, $1 \leq |r| \leq 2$



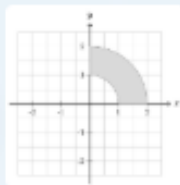
A

9%



B

9%



C

82%



D

0%



$\pi/2 \approx 90^\circ$
 $1 \leq r \leq 2$
 $1 \leq -r \leq 2$
 $-2 \leq r \leq -1$
 $r > 0$
 $r < 0$





$r = 0 \rightarrow$ represent origin $(0,0)$

When converting cartesian equation $(x - 5)^2 + y^2 = 25$ to polar equation, I write

$$x^2 - 10x + 25 + y^2 = 25 \implies x^2 + y^2 = 10x \implies r^2 = 10r \cos \theta \implies r = \pm \sqrt{10r \cos \theta}$$

. What do you think about this argument?

$$r^2 = x^2 + y^2$$



$$r = 10 \cos \theta$$

Sounds good to me.

It is flawed.

It is flawed but I cannot find a reason for that.

It is technically correct but it gives no information on the values of

r .



Responses Hidden

Edit

Show Choices



Show Responses



Lock



Show Correctness



Implicit defined parametrization

x and y are functions of t

$$\frac{d}{dt} x = x'(t) \quad , \quad \frac{dy}{dt} = y'(t)$$

$$\boxed{\frac{d}{dt} x = 0 \quad , \quad \frac{d}{dt} y = 0} \quad \text{Wrong}$$

Always look for product rule / chain rule.

Ex 10.2 Q19

$$x = t^3 + t, \quad y + 2t^3 = 2x + t^2, \quad t = 1$$

Aim: $\boxed{\frac{dy}{dx} \big|_{t=1}}$ $\rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ ✓

- ① Find $\frac{dy}{dt}, \frac{dx}{dt}$
 - ② Find $\frac{dy}{dx}$ (not final value)
 - ③ Find values of x, y when $t = 1$
 - ④ $\frac{dy}{dx} \big|_{t=1}$
- # optimal / depends on qns

$$\frac{dx}{dt}:$$

$$x = t^3 + t$$

$$\frac{dx}{dt} = 3t^2 + 1$$

$$\frac{dy}{dt}:$$

$$y + 2t^3 = 2x + t^2$$

Diff. with respect to t :

$$\frac{dy}{dt} + \underline{6t^2} = 2 \frac{dx}{dt} + 2t$$

$$\frac{dy}{dt} = 2(3t^2 + 1) + 2t - 6t^2 = \underline{\underline{2 + 2t}}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2+2t}{3t^2+1}$$

$$\left. \frac{dy}{dx} \right|_{t=1} = \frac{2+2(1)}{3(1)^2+1} = 1$$

Q17 (sketch) $\underline{y\sqrt{t+1}} + \underline{2t\sqrt{y}} = 4$

$$\underline{\frac{dy}{dt}} \sqrt{t+1} + y \cdot \frac{1}{2\sqrt{t+1}} + 2\sqrt{y} + 2t \frac{1}{2\sqrt{y}} \underline{\frac{dy}{dt}} = 0$$

Cartesian $\rightarrow$ Polar

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\underline{x^2 + y^2 = r^2}$$

$$\sin 2\theta$$

$$= 2 \cos \theta \sin \theta$$



$$r^2 \cdot \frac{1}{2} \sin 2\theta = 2$$

$$\boxed{r^2 \sin 2\theta = 4}$$



Correct

Ex. 10.3 Q60 $xy = 2$

$$\Rightarrow r \cos \theta \cdot r \sin \theta = 2$$

$$\Rightarrow \boxed{r^2 \cos \theta \sin \theta = 2}$$

$$\Rightarrow \boxed{r^2 = \frac{2}{\cos \theta \sin \theta}} \quad \text{[WRONG]} \quad \text{Can be zero}$$

$$\theta = 0$$

$$\sin \theta = 0$$

$$\cos \theta = 1$$